

02 - IVPs; direction fields

1. **Daily Problem.** Last class we found that the functions $y(x) = Ce^x$ (for any real number C) are solutions of the ODE $\frac{dy}{dx} = y$. Now, find as many solutions as you can to the ODEs $\frac{dy}{dx} = 5y$ and $\frac{dy}{dx} = 5x$.



$$y = \frac{5}{2}x^2 + C$$



$$y = Ce^{5x}$$

A general solution of an ODE is

a representation of all solns of that ODE

Let's generalize a principle from the previous problem: the general solution of an ODE of the type

is given by:

An initial value problem (IVP) is

A problem in which

we must find a general soln for an ODE given ≥ 1 ICs

2. Solve the Initial Value Problem (IVP) $\frac{dy}{dx} = 5y$ with $y(-2) = 1$. How many solutions did you find?

general soln $y = Ce^{5x}$
 substitute in $x = -2$

$$1 = Ce^{-10} \quad C \text{ must be } e^{10}$$

exactly 1 soln, 1st degree ODE
 w/ 1 IC

3. Find the general solution to the differential equation $\frac{dy}{dt} = t^2 + 5$.

$$\frac{t^3}{3} + 5t + C$$

v.a Integration

4. (a) Solve the differential equation $\frac{d}{dt}(ty) = 5t$ for the unknown function $y(t)$.

- (b) Why does your strategy not work to solve $\frac{d}{dt}(ty) = 5y$ instead?

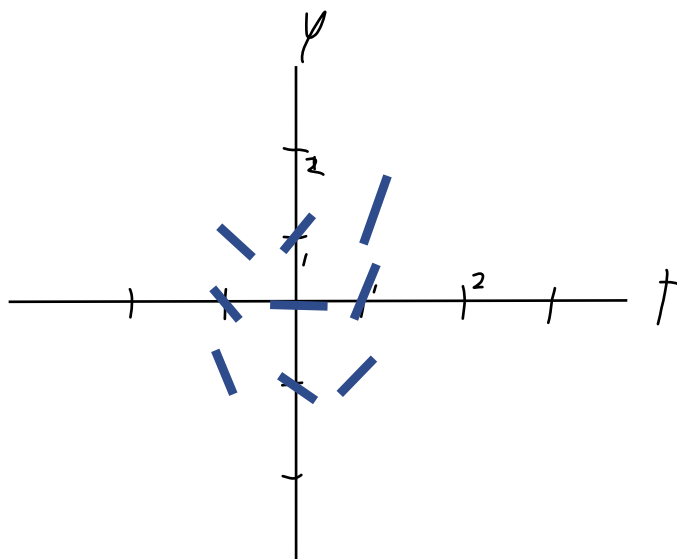
5. **Direction fields.** Consider the differential equation $\frac{dy}{dt} = 2t + y$. You can think of $\frac{dy}{dt}$ as depending on both t and y . For example, when $t = 1$ and $y = 3$ then $\frac{dy}{dt} = 2 \times 1 + 3 = 5$.

(a) ~~10 DE~~ Fill in the following table accordingly:

t	y	$dy/dt = 2t + y$
-1	-1	-3
0	-1	-1
1	-1	1
-1	0	-2
0	0	0
1	0	2
-1	1	-1
0	1	1
1	1	3

(b) How would you describe all the points (t, y) for which $\frac{dy}{dt} = 0$?

(c) Draw the direction field for this ODE.



Why construct a direction field?

To sketch a few representative solutions, and have a rough sense of what the solutions look like.

How to construct a direction field?

1. Draw a grid on the t, y plane.
2. At each grid point, draw short line segments that have slope dy/dt given by the ODE.

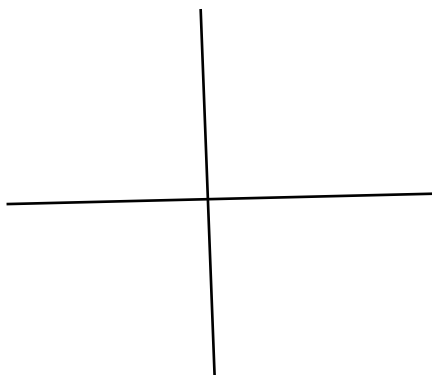
6. To simplify our work, and to give us better intuition for the solutions, we use isoclines. An isocline of an ODE $\frac{dy}{dt} = f(t,y)$ is a set of points (t,y) where the slope (cline) is always the same (iso). Which isocline did we consider in the last problem? (We call this special kind of an isocline a nullcline.)



7. Draw the direction fields for the following differential equations, with a few representative solutions.

(a) $\frac{dy}{dt} = 1$.

(b) $\frac{dy}{dt} = -y$.



(c) $\frac{dy}{dt} = -t$.

(d) **Daily Problem.** Please try this question in preparation for next class. $\frac{dy}{dt} = -\frac{t}{y}$.

8. Let's reflect back on some shortcuts we learned from the previous problem:

- If you have a differential equation of the form $\frac{dy}{dt} = f(t)$ (that is, $\frac{dy}{dt}$ depends only on t , not on y ; an example is $\frac{dy}{dt} = t^2 + 1$), what can you say about its direction field and isoclines? How do different solutions of the differential equation relate to each other?
 - If you have a differential equation of the form $\frac{dy}{dt} = f(y)$ (that is, $\frac{dy}{dt}$ depends only on y , not on t), what can you say about its direction field and isoclines? How do different solutions of the differential equation relate to each other?
9. For each direction field in #7, can you guess the general solution from looking at the direction field? Check your answer by plugging back in the ODE! (In a week or two, you should be able to solve each ODE analytically, so come back and do that, and make sure you get the same answers!)

10. Here are four differential equations. Match each with a direction field below.

(a) $\frac{dS}{dt} = 2 - S/50$. (b) $\frac{dS}{dt} = 50(\frac{100}{S} - 1)$. (c) $\frac{dS}{dt} = 2 - \frac{2S}{100-t}$. (d) $\frac{dS}{dt} = \frac{200-2S}{100-t}$.

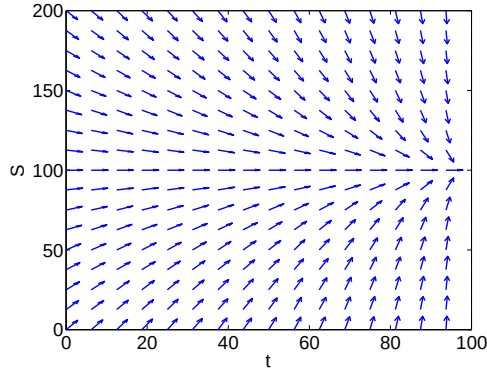


Figure 1: (A)

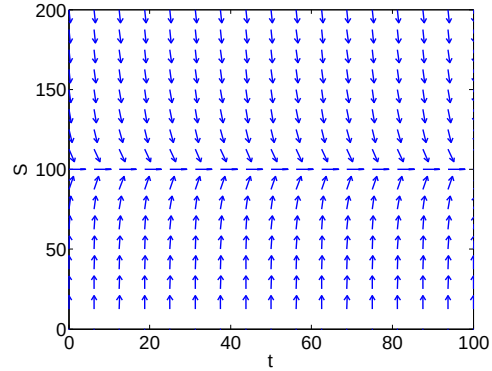


Figure 3: (C)

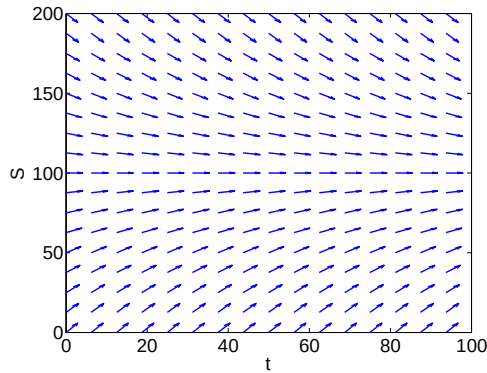


Figure 2: (B)

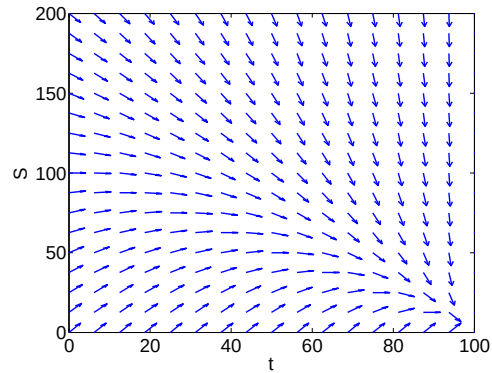


Figure 4: (D)

11. For each of the direction fields from the previous question, draw qualitatively correct solutions for the IVPs with $S(0) = 0$, $S(0) = 100$ and $S(0) = 200$. For each direction field, say whether the behavior of solutions as t becomes very large depends on the initial condition.

12. (a) Draw the direction field for the differential equation $\frac{dy}{dt} = 1 - y$. Sketch two solutions.

(b) Which of the following is a solution to $\frac{dy}{dt} = 1 - y$? (Can you guess from the direction field before trying them all out?)

i. $y = Ce^{-t}$

iii. $y = Ce^{-t} - 1$

v. $y = Ce^t + 1$

ii. $y = Ce^t - t$

iv. $y = Ce^{-t} + 1$